ANTONIO RAPUANO

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SUMMARY

I'm an engineer driven by curiosity and a strong desire to turn theory into practical systems. I enjoy working across the entire development pipeline – from modeling and simulation to implementation and testing. I'm looking for a role where I can keep learning and contributing to impactful technologies, especially in robotics and autonomous systems.

EXPERIENCE (Details and media at ntonioa.github.io/about)

Autopilot Control Systems Engineer — Fall 2024 - Present
Sapienza Flight Team (sasa-aerospace.it/flight-team)
Highlights: Control architecture design • Implementation on custom hardware • Testing and validation

Race Timing Specialist and Team Lead — Fall 2018 - Present
ICRON (icron.it)
Highlights: Race logistics and technical setup • Start/finish line operations • Ranking and award management

EDUCATION (Details and media at ntonioa.github.io/about)

M. Sc. in Control Engineering — Fall 2022 - Summer 2025
Università degli Studi di Roma "La Sapienza", Rome, Italy — Final grade: 110/110 with honors
Thesis: Nonlinear predictive control of the continuum and hybrid dynamics of a suspended deformable cable for aerial pick and place (Supervisor: Prof. Antonio Franchi)
Relevant coursework: Analysis, modeling and control of nonlinear, hybrid, multivariable, networked systems • Basic and advanced robotics • Estimation • Optimal control • Machine learning • Robust control

B. Sc. in Electronic Engineering for Automation and Telecommunications — Fall 2018 - Winter 2022
Università degli Studi del Sannio, Benevento, Italy — Final grade: 110/110 with honors
Thesis: Implementation on FPGA of an electronic circuit controlling a mechanical arm (Supervisor: Prof. Marco Pisco)
Relevant coursework: Analog and digital electronics • Analysis, modeling and control of linear systems • Digital design • Electronic measurements • Signal processing • Telecommunications

ACADEMIC PROJECTS (Details and media at ntonioa.github.io/projects)

- Control of underactuated robots via input-constrained receding-horizon DDP — Winter 2024 - Fall 2024
- Fault-tolerant formation control of passive multi-agent systems using energy tanks — Summer 2024
- Decentralized PEV charging control based on a subgradient method for MILPs — Winter 2023 - Spring 2024
- Modeling and control of the Mars helicopter 'Ingenuity' — Fall 2023 - Spring 2024
- Control of a video-game race car using a convolutional neural network — Winter 2024
- Self-balancing two-wheeled robot — Fall 2023 - Winter 2024
- Signal acquisition through an A/D converter — Spring 2021

SKILLS

Software and tools: Arduino • Blender • Fusion 360 • LTSpice • Office • Quartus Prime • Simulink • Stateflow

Programming and formatting: C/C++ • Java • HTML/CSS • LaTeX • MATLAB • Python • Verilog

Hardware: 3D printing • Circuit board soldering • Measuring

Personal skills: Attention to detail • Autonomy and initiative • Critical thinking • Curiosity and continuous learning • Problem-solving attitude • Project planning • Technical writing • Time management

LANGUAGES

Italian, native

English, fluent
Cambridge English C1 Advanced - Score 207 (Grade A, CEFR Level C2) — Summer 2022